

A Fast Electrothermal Response Analysis Method for Interconnect Circuits With Multiexcitations in Multilayer Packages

Yang Liu^{1b}, Dehang Wu^{1b}, *Student Member, IEEE*, Zhifei Xu, *Member, IEEE*,
Xiuqin Chu^{2b}, *Senior Member, IEEE*, Jun Wang^{1b}, *Member, IEEE*, and Kai-Da Xu^{1b}, *Senior Member, IEEE*

Abstract—The design of package interconnects is increasingly vital as integrated circuits (ICs) continue to shrink in size and increase in density. However, reducing interconnect size may lead to various issues such as increased temperature, resistivity, and decreased reliability. Accurate estimation of the electrothermal response of interconnect structures is crucial in guiding circuit design. In this article, an efficient method for transient thermal analysis has been proposed for interconnect circuits with complex stacking structures and multiexcitations. This method utilizes the tensorial analysis network (TAN) in combination with the Cauer I RC network (i.e., RC-TAN) to analyze the electrothermal sensitivity of package interconnects. Then, the main parameters affecting the electrothermal response and the optimal interconnect layer stacking order can be derived to guide the design of package interconnects. The effectiveness of this method has been verified by comparing the results with those obtained through finite element method (FEM) simulation. It indicates that the proposed method can save up to 99.8% of the computation time compared to FEM. Furthermore, the maximum temperature error is only 1.7% and the steady-state error is below 1%, highlighting the method's high accuracy.

Index Terms—Cauer I RC network, electrothermal response, package interconnect, tensorial analysis network (TAN).

I. INTRODUCTION

AS THE semiconductor industry continues to develop, the traditional framework of Moore's Law is no longer adequate to describe the integrated circuit (IC) processes. The principle of equal scaling down of feature size is no

longer fully applicable in future IC development. In the current industry landscape, most IC manufacturing experiences a slowdown in technology node variation and wafer size changes despite increasing capital expenditures. This indicates that the industry is undergoing a shift toward a different paradigm for IC development.

To maintain their leadership in producing ICs with smaller size, lower cost and higher performance, manufacturers can integrate more advanced chip packaging technologies throughout the manufacturing process, such as 2.5-D and 3-D IC technologies [1], [2], [3]. These packaging technologies are proven to offer high-quality chip connections and lower power consumption compared to traditional packaging methods. Although most of these technologies are still in development, they hold great promise for the future of IC manufacturing [4], [5]. Fig. 1 displays these advanced packaging technologies and multilayer interconnects. It has become essential to implement package stacking technology to enhance overall performance without upgrading the chip process. This necessitates the use of more interconnects to facilitate signal transmission between chips in the same area. The traditional approach involves reducing interconnect size to increase the number of interconnects within a limited area. However, process constraints impose limits on the size of interconnects. Therefore, a multilayer structure is required by stacking several interconnects vertically to accommodate more signal lines within the same area [6], [7], but this interconnect structure may also lead to new thermal problems.

As the width of interconnect lines decreases and the number of signal layers increases, the power consumption on interconnects increases, leading to a rise in interconnect temperature. This increase in temperature, in turn, leads to a rise in resistivity of metal interconnects and power consumption, thereby progressively reducing circuit performance and reliability. This issue has made interconnects a significant bottleneck in the performance of ICs [8], [9]. Consequently, researchers have started to focus more attention on analyzing the electrothermal response.

The investigation of electrothermal problem in the literature can be categorized into three aspects: experimental method for obtaining data to guide design [10], numerical calculation method for predicting the electrothermal response of interconnects [11], [12], [13], [14], [15], [16], [17], [18], and

Manuscript received 9 April 2023; revised 2 July 2023; accepted 16 July 2023. Date of publication 19 July 2023; date of current version 17 August 2023. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFC2203903, in part by the National Natural Science Foundation of China under Grant 62171338 and Grant 61901331, in part by the Natural Science Basic Research Program of Shaanxi Province under Grant 2021JM-132, and in part by the Fundamental Research Funds for the Central Universities under Grant XJS210203. Recommended for publication by Associate Editor P. Franzon upon evaluation of reviewers' comments. (*Corresponding authors: Yang Liu; Kai-Da Xu.*)

Yang Liu, Dehang Wu, Xiuqin Chu, and Jun Wang are with the Key Laboratory of High Speed Circuit Design and EMC of Ministry of Education, School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: liuyang@mail.xidian.edu.cn).

Zhifei Xu is with the High Speed Platform Department, Ningbo Detool Technology Company Ltd., Ningbo 315000, China.

Kai-Da Xu is with the School of Information and Communications Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: kaidaxu@ieee.org).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCPMT.2023.3296960>.

Digital Object Identifier 10.1109/TCPMT.2023.3296960

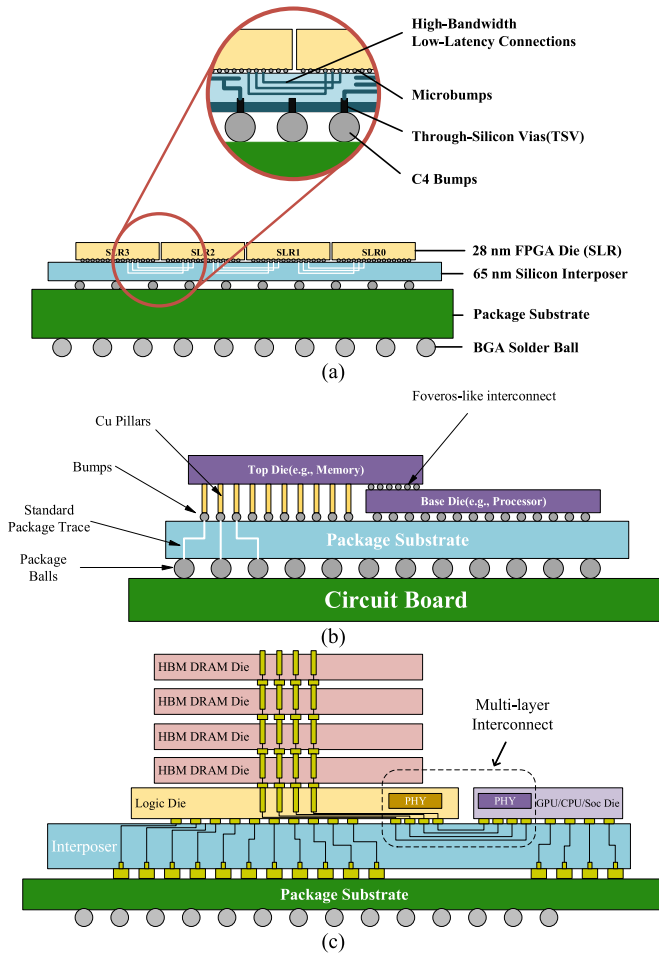


Fig. 1. Chip stacking technology schematic. (a) 2.5-D stacking schematic. (b) 3-D stacking schematic. (c) Multilayer interconnects in chiplet schematic.

method of the machine learning and deep learning for fast thermal analysis of interconnections [19], [20], [21], [22], [23]. The numerical calculation method can be further divided into two types. One type is that using computational methods [e.g., finite difference time domain (FDTD), finite element method (FEM), boundary element method (BEM), and discontinuous Galerkin time domain (DGTD)] to analyze the thermal field and solve the heat conduction equation, but it is very time-consuming and needs high-performance hardware equipment for computing. The other type is using the equivalent thermal circuit method to calculate the temperature response of interconnects. This method involves extracting the thermal capacitance and thermal resistance of the interconnect structure to construct an equivalent thermal circuit. Although this method is more efficient and faster, its accuracy needs to be improved [24], [25], [26], [27]. Moreover, it cannot calculate the thermal coupling between multiple layers of signal lines and the electrothermal response for multiple excitation cases simultaneously. For the method of machine learning and deep learning, it is achieved by importing training parameters such as the power map distributions on the heating plane and the interconnect size, which is highly adaptive for various scenarios, but its accuracy is significantly dependent on data amount and computational resources.

The tensorial analysis of network (TAN) [28] is a topology-based mathematical method that was initially developed in the field of mechanics. Kron [29] applied it to electrical circuits in 1942. In recent decades, researchers have found that the TAN is well-suited for the multiphysics field analysis. Olivier Maurice extended Kron's approach to the field of electromagnetic compatibility [30] [31] and explored the possibilities of the TAN approach in thermal analysis [32].

In this article, we combine the tensorial analysis network (TAN) with the Cauer I RC network to form an RC-TAN method for analyzing the electrothermal sensitivity of package interconnects. The primary contributions of this research can be summarized as follows.

- 1) The TAN approach is applied to the electrothermal analysis of package interconnects, expanding its application area. With the advantage of TAN adapting to multiple physical fields, the joint simulation of electricity, heat and magnetism can be realized after the multiphysics TAN is perfected.
- 2) The proposed RC-TAN method can save more than 99% of the calculation time compared to widely used thermal field calculation methods.
- 3) The RC-TAN approach gives a general equivalent thermal circuit model for multilayer interconnected structures based on the equivalent thermal circuit method. This model can improve the thermal coupling analysis of multilayer signal lines and add support for parallel operation of multiple excitations.
- 4) Using the RC-TAN approach's extremely high computational efficiency and support for multilayer thermal coupling analysis, the electrothermal sensitivity of the interconnect structure dimensions and the influence of the layer stacking order on the electrothermal response of the interconnect structure are studied and analyzed to guide the design of the interconnect structure.

The article is organized as follows. Section II presents the main theoretical and simulation calculation using the RC-TAN method. In Section III, a generalized equivalent thermal circuit model of a multilayer and multiexcitation interconnect structure is presented. Section IV provides the results of the temperature response of the interconnects under electrostatic discharge (ESD) [33] excitation, which are calculated by the RC-TAN method and compared with the simulation results by the ANSYS software. In Section V, the electrothermal sensitivity of the package interconnect structure is analyzed. Finally, concluding remarks are made in Section VI.

II. THEORETICAL FRAMEWORK OF THE RC-TAN METHOD

A. Interconnect Electrothermal RC Model

In a printed circuit board (PCB), interconnect lines and each layer of dielectric are typically distributed evenly. As a signal propagates through the interconnect lines, power is also evenly distributed across them. Heat in the interconnect structure is mainly conducted vertically. Therefore, in thermal analysis, interconnects are divided horizontally into many volume elements. By calculating the temperature response of a volume

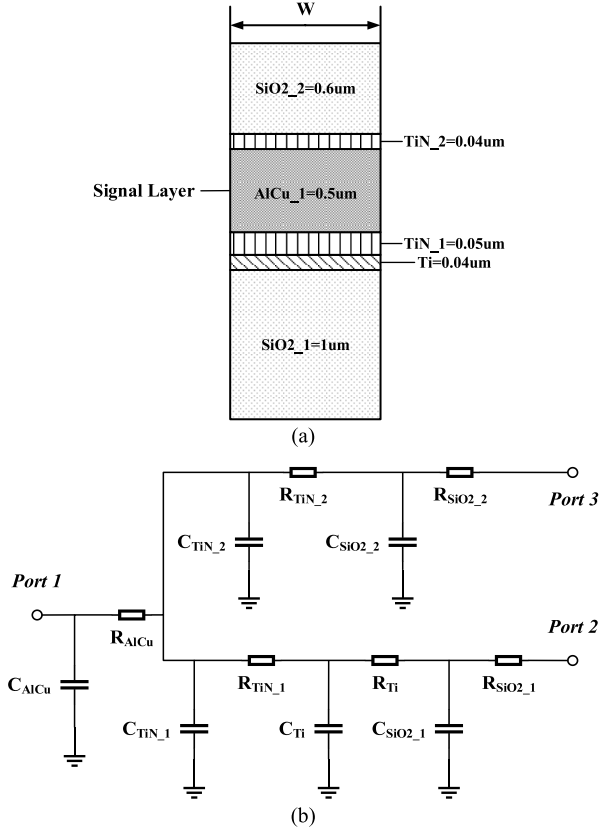


Fig. 2. Interconnect of single signal layer. (a) Interconnect stacking structure. (b) Interconnected electrothermal RC model.

element, the temperature response of the entire interconnect line can be obtained. For the interconnect structure shown in Fig. 2(a), a first-order Cauer I RC network is used to model each layer. The thermal resistance and thermal capacitance in the equivalent circuit denoted as “ R_T ” and “ C_T ” can be obtained using the following equations:

$$R_T = \frac{h}{K \cdot L \cdot W} \quad (1)$$

$$C_T = \rho \cdot c_p \cdot h \cdot L \cdot W \quad (2)$$

where h , L , and W correspond to the high, length and width of the solid with unit of μm , ρ is density of the solid with unit of kg/m^3 , K is the thermal conductivity of the solid with unit of $\text{W}/^\circ\text{C} \cdot \text{m}$, and c_p is the specific heat of the solid with unit of $\text{J}/^\circ\text{C} \cdot \text{kg}$.

In the equivalent thermal circuit shown in Fig. 2(b), Port1 is connected to the signal layer for input signal excitation, while Ports 2 and 3 are connected to the outermost layer as the input ports of ambient temperature. The metal conductor layer, which forms the interconnects, has good thermal conductivity and large thermal capacity, resulting in a fast rate of thermal conduction that can be accurately modeled using only the first-order RC thermal circuit model. On the other hand, the dielectric layer, which has a low thermal conductivity and is often thick, results in a slow rate of thermal conduction that cannot be accurately modeled using only the first-order model. As a result, there is a certain error in the analytical results of the interconnect temperature response. To address this error, a multiorder RC thermal circuit model is used to

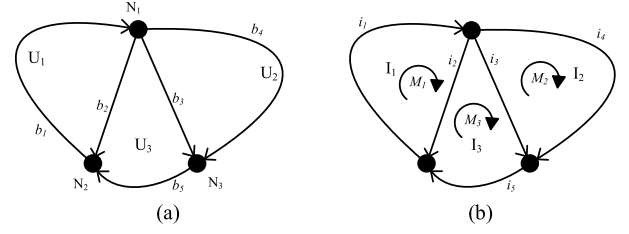


Fig. 3. Topological expression of different spaces. (a) Mesh space. (b) Branch and nodal space.

model the conductor layer of the interconnects. The larger thermal resistance of the dielectric layer is distributed into the multiorder RC to improve the accuracy of the RC-TAN method theoretically.

B. TAN Method

The TAN method involves transforming the topology of a circuit into a graph comprising of nodes, branches, and meshes. Fig. 3 illustrates a mesh space that has been derived from a third-order circuit using this method. Within the mesh space, the Kirchhoff's law is applied to determine the relationship between the currents and voltages of the various meshes and branches. Notably, the number of meshes in the mesh space is typically lower than the number of branches in the original circuit topology. As a result, computing the response in the mesh space is faster and more efficient. The source E_n ($n = 1, 2, 3$) [34] can be obtained by analyzing the vectors of different spaces as

$$\begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix} \cdot \begin{bmatrix} I^1 \\ I^2 \\ I^3 \end{bmatrix} \quad (3)$$

where the third-order matrix Z is the impedance matrix, I^n ($n = 1, 2, 3$) is the current of the corresponding mesh.

C. Proposed RC-TAN Method

To implement the RC-TAN method, the topology of the thermal RC model is transformed into a graph consisting of nodes, branches, and meshes. The topology of the thermal RC model is composed of the classical schematic of the Y-tree. Fig. 4 shows the classical schematic of Y-tree with input port M_1 as well as output ports M_2 and M_3 . According to the circuit shown in Fig. 4(a), the equivalent graph shown in Fig. 4(b) can be obtained. Then, the impedance matrix for the topology of the thermal path can be calculated. The mesh impedance given in Fig. 4 can be written as

$$Z_{mn}(s) = \begin{bmatrix} Z_{11} & 0 & 0 & \frac{1}{sC_1} & \frac{1}{sC_1} \\ 0 & Z_{22} & 0 & \frac{1}{sC_1} & 0 \\ 0 & 0 & Z_{33} & 0 & \frac{1}{sC_1} \\ \frac{1}{sC_1} & \frac{1}{sC_1} & 0 & Z_{44} & R_1 + \frac{1}{sC_1} \\ \frac{1}{sC_1} & 0 & \frac{1}{sC_1} & R_1 + \frac{1}{sC_1} & Z_{55} \end{bmatrix} \quad (4)$$

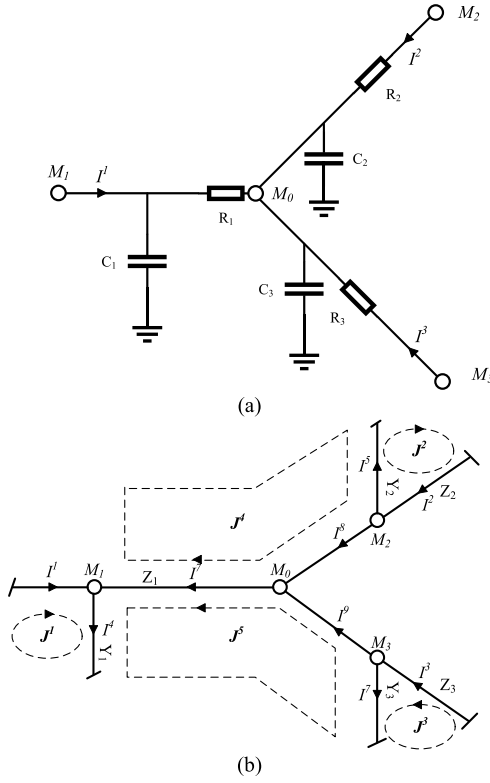


Fig. 4. Y-tree network. (a) Y-tree classical circuit. (b) Y-tree equivalent graph.

where $Z_{11} = (1/sC_1)$, $Z_{22} = R_2 + (1/sC_2)$, $Z_{33} = R_3 + (1/sC_3)$, $Z_{44} = R_1 + (1/sC_1) + (1/sC_2)$, $Z_{55} = R_1 + (1/sC_1) + (1/sC_3)$.

During the calculation of the temperature response, excitation is input on ports and the focus is paid on the temperature of the layer where the signal is transmitted. Therefore, the impedance matrix of the complete circuit needs to be transformed into a port impedance matrix with lower order, which helps us better focus on the temperature response of the target layer and significantly improve the computation efficiency by reducing invalid operations [34].

As a matter of a fact, the mesh impedance $Z_{mn}(s)$ can be subdivided, as illustrated in the following:

$$Z_{mn} = \begin{bmatrix} Z_a & Z_b \\ Z_c & Z_d \end{bmatrix}. \quad (5)$$

Four sub-matrices of Z_{mn} are denoted as:

- 1) Z_a having size $P \times P$;
- 2) Z_b having size $P \times (M - P)$;
- 3) Z_c having size $(M - P) \times P$;
- 4) Z_d having size $(M - P) \times (M - P)$.

P and M represent the numbers of ports and meshes.

Then we obtain the impedance matrix as follows:

$$[Z] = [Z_a] - [Z_b] \times [Z_d]^{-1} \times [Z_c]. \quad (6)$$

The port impedance matrix contains the impedance information of all ports. Therefore, it is possible to calculate the temperature response of each port for multiple excitations at the same time. After obtaining the port impedance matrix, the temperature response of multiple ports can be calculated by

the following equation:

$$\begin{bmatrix} T_1(s) \\ T_2(s) \\ \vdots \\ T_n(s) \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1n} \\ Z_{21} & Z_{22} & \cdots & Z_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{n1} & Z_{n2} & \cdots & Z_{nn} \end{bmatrix} \cdot \begin{bmatrix} P_1(s) \\ P_2(s) \\ \vdots \\ P_n(s) \end{bmatrix} \quad (7)$$

where the $T_n(s)$ and $P_n(s)$ represent temperature response and power on interconnect line in frequency domain, respectively.

Equation (7) gives the temperature response in the frequency domain, which can be transformed into the time domain by using Laplace inverse transform. However, the complexity of the frequency domain expression is exponentially related to the size of the matrix, making it challenging to perform Laplace inverse transform of the analytic equation for complex interconnect structures in cascade. Conventionally, reducing the order of the higher-order analytic formula is done to ease computational difficulty, but this method comes at the expense of accuracy. As the structure becomes more complex, the reduction in accuracy becomes more significant. Therefore, this work employs a numerical inverse transformation method [35] to derive the exact solution of the temperature response in the frequency domain and then perform numerical inverse transformation to obtain the exact time-domain temperature response. The derivation of the numerical inverse Laplace transform method can be found in Appendix.

III. MULTILAYER AND MULTIEXCITATION THERMAL CIRCUIT ANALYSIS

As known from Section II, the equivalent thermal model of single-layer interconnects is the classical Y-tree, with a laminated structure of the middle signal layer and two dielectric layers, while the laminated structure of multilayer interconnect is an alternating stack of dielectric and signal layers. Therefore, the multilayer interconnect structures can be considered as a cascade of multiple single-layer interconnect structures, and its equivalent thermal model becomes a cascade of multiple Y-tree models. Based on this theory, we give the equivalent thermal circuit model shown in Fig. 5(b) for the multilayer interconnect structure shown in Fig. 5(a) in its generic form, where Port 1 to Port n are the input ports of the signal layer, Port $n + 1$ and Port $n + 2$ are the ports of the dielectric layer in contact with air.

When an N layers interconnect structure has S signal layers, the number of ports is $P = S + 2$ and the number of meshes is $M = N + S - 1$. If the interconnect structure has not only multiple ports but also complex cascade, it becomes tedious to obtain the impedance matrix for the complete thermal circuit of the entire structure. In order to make the impedance extraction of the equivalent thermal circuit easier, a fast method to obtain the impedance matrix is given based on the generic thermal circuit model with multiple layers and multiple ports proposed above.

The generic electrothermal RC model of the multilayer interconnect shown in Fig. 5(b) can be obtained by cascading the Y-tree, and therefore it has a certain regularity. Its cascade structure is a cycle of dielectric layer-signal layer-dielectric layer in general. Based on this property, we can split its

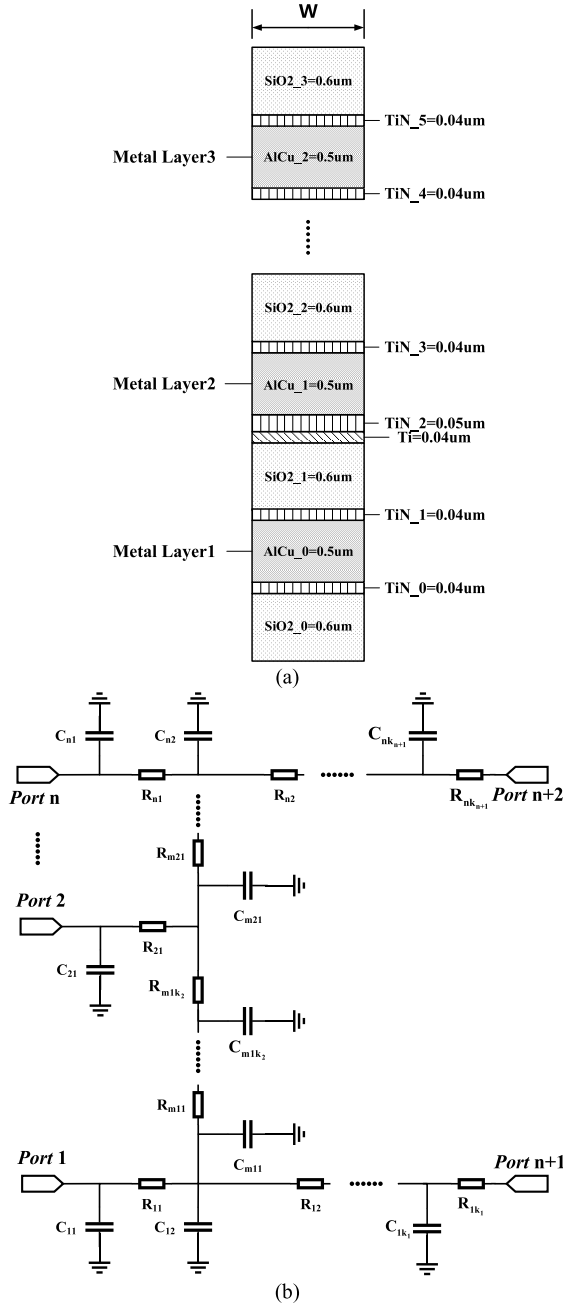


Fig. 5. Multilayer interconnect electrothermal model. (a) Multilayer interconnect cascade schematic. (b) Multilayer interconnect equivalent thermal circuit.

thermal circuit model. Taking the interconnect structure of dual signal layers as an example, we can split it as shown in Fig. 6(b).

The rules for splitting are as follows:

- 1) First signal layer to first dielectric layer;
- 2) The last signal layer to the last dielectric layer;
- 3) Signal layer between the two dielectric layers;
- 4) Dielectric layer between the two signal layers.

After splitting the thermal circuit according to the above rules and using the TAN method to quickly obtain the impedance matrix of each sub-circuit, the following equation can be used to cascade the signal layer with the dielectric in

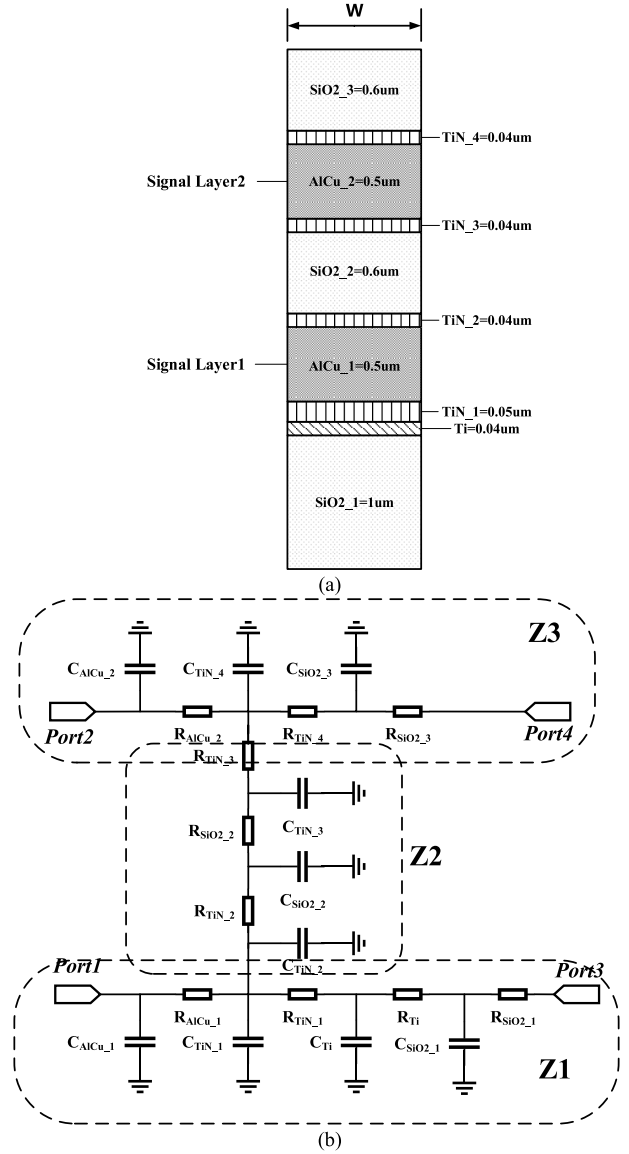


Fig. 6. Two-signal layer interconnect. (a) Two-signal layer interconnect stacking structure. (b) Electrothermal model of (a).

sequence to obtain the impedance matrix after cascading

$$Z_{mn} = \begin{bmatrix} \begin{matrix} Z1_{11} & Z1_{12} & \cdots & Z1_{1n} \\ Z1_{21} & Z1_{22} & \cdots & Z1_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ Z1_{n1} & Z1_{n2} & \cdots & Z1_{nn} \end{matrix} & \begin{matrix} -Z1_{11} \\ Z_c \\ \vdots \\ -Z1_{11} \end{matrix} \\ -Z1_{11} & Z_c \end{bmatrix} \begin{matrix} \text{Impedance matrix of} \\ \text{sub-circuit 1} \end{matrix}$$

$$\begin{matrix} \text{Impedance matrix of} \\ \text{sub-circuit 2} \end{matrix} \begin{bmatrix} \begin{matrix} Z2_{11} & Z2_{12} & \cdots & Z2_{1m} \\ Z2_{21} & Z2_{22} & \cdots & Z2_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ Z2_{m1} & Z2_{m2} & \cdots & Z2_{mm} \end{matrix} \end{bmatrix}$$

(8)

where Z_c is the coupling impedance between two sub-circuits.

Take the interconnect structure of two signal layers shown in Fig. 6(a) as an example. According to the TAN method we

can easily obtain the impedance matrix of each sub-circuit, as in (9)–(11), shown at the bottom of the page.

The impedance matrix of the complete thermal circuit can be obtained by sequential cascading the above three impedance matrices using the (8). The impedance matrix for a complete thermal circuit is as in (12), shown at the bottom of the page, where

$$\begin{aligned} Z_{C1} &= R_{AlCu1} + \frac{1}{sC_{AlCu1}} \\ Z_{C2} &= R_{AlCu2} + \frac{1}{sC_{AlCu2}}. \end{aligned} \quad (13)$$

IV. SIMULATION RESULTS

For the interconnect structure shown in Fig. 2(a) with $W = 1.1 \mu\text{m}$, $L = 100 \mu\text{m}$, the thermal parameters of each material are given in Table I. Substituting the size and thermal properties of solids into (1) and (2) to obtain the equivalent thermal capacity and equivalent thermal resistance of each layer.

The first-order RC equivalent thermal circuit model has met the accuracy requirements of the metal conductor layer, and the equivalent thermal resistance of the metal conductor layer is 20.55Ω . Based on this information, we can learn that the accuracy of the temperature response analysis results

$$Z1 = \begin{bmatrix} \frac{1}{sC_{AlCu1}} & -\frac{1}{sC_{AlCu1}} & & & \\ -\frac{1}{sC_{AlCu1}} & R_{AlCu1} + \frac{1}{sC_{AlCu1}} + \frac{1}{sC_{TiN1}} & -\frac{1}{sC_{TiN1}} & & \\ & -\frac{1}{sC_{TiN1}} & R_{TiN1} + \frac{1}{sC_{TiN1}} + \frac{1}{sC_{Ti}} & -\frac{1}{sC_{Ti}} & \\ & & -\frac{1}{sC_{Ti}} & R_{Ti} + \frac{1}{sC_{Ti}} + \frac{1}{sC_{SiO2_1}} & -\frac{1}{sC_{SiO2_1}} \\ & & & -\frac{1}{sC_{SiO2_1}} & R_{Ti} + \frac{1}{sC_{SiO2_1}} \end{bmatrix} \quad (9)$$

$$Z2 = \begin{bmatrix} R_{AlCu1} + \frac{1}{sC_{AlCu1}} + \frac{1}{sC_{TiN2}} & -\frac{1}{sC_{TiN2}} & & & \\ -\frac{1}{sC_{TiN2}} & R_{TiN2} + \frac{1}{sC_{TiN2}} + \frac{1}{sC_{SiO2_2}} & -\frac{1}{sC_{SiO2_2}} & & \\ & -\frac{1}{sC_{SiO2_2}} & R_{SiO2_2} + \frac{1}{sC_{SiO2_2}} + \frac{1}{sC_{TiN3}} & -\frac{1}{sC_{TiN3}} & \\ & & -\frac{1}{sC_{TiN3}} & R_{TiN3} + R_{AlCu2} + \frac{1}{sC_{TiN3}} + \frac{1}{sC_{AlCu2}} & \\ & & & -\frac{1}{sC_{AlCu2}} & \end{bmatrix} \quad (10)$$

$$Z3 = \begin{bmatrix} \frac{1}{sC_{AlCu2}} & -\frac{1}{sC_{AlCu2}} & & & \\ -\frac{1}{sC_{AlCu2}} & R_{AlCu2} + \frac{1}{sC_{AlCu2}} + \frac{1}{sC_{TiN4}} & -\frac{1}{sC_{TiN4}} & & \\ & -\frac{1}{sC_{TiN4}} & R_{TiN4} + \frac{1}{sC_{TiN4}} + \frac{1}{sC_{SiO2_3}} & -\frac{1}{sC_{SiO2_3}} & \\ & & -\frac{1}{sC_{SiO2_3}} & R_{SiO2_2} + \frac{1}{sC_{SiO2_3}} & \end{bmatrix} \quad (11)$$

$$Z_{mn} = \begin{bmatrix} Z1_{11} & Z1_{12} & \cdots & Z1_{15} & -Z1_{11} & & & & \\ Z1_{21} & Z1_{22} & \cdots & Z1_{25} & Z_{C1} & & & & \\ \vdots & \vdots & \ddots & \vdots & & & & & \\ Z1_{51} & Z1_{52} & \cdots & Z1_{55} & & & & & \\ -Z1_{11} & Z_{C1} & & & Z2_{11} & Z2_{12} & \cdots & Z2_{14} & \\ & & & & Z2_{21} & Z2_{22} & \cdots & Z2_{24} & \\ & & & & \vdots & \vdots & \ddots & \vdots & \\ & & & & Z2_{41} & Z2_{42} & \cdots & Z2_{44} & \\ & & & & & & & -Z3_{11} & Z_{C2} & & \\ & & & & & & & -Z3_{11} & Z3_{12} & \cdots & Z3_{14} \\ & & & & & & & Z_{C2} & Z3_{21} & Z3_{22} & \cdots & Z3_{24} \\ & & & & & & & & \vdots & \vdots & \ddots & \vdots \\ & & & & & & & & Z3_{41} & Z3_{42} & \cdots & Z3_{44} \end{bmatrix} \quad (12)$$

TABLE I
THERMAL PROPERTIES OF A FEW SOLIDS

Material	K (W / °C · m)	ρ (kg / m ³)	c_p (J / °C · kg)
AlCu	221.2	2700	899.6
TiN	25	4500	519
Ti	21.9	4550	460
SiO2	1.4	2260	230

can be guaranteed when the thermal resistance of a single RC thermal circuit model unit is about 20 Ω . In this article, the total thermal resistance of the dielectric layer of the interconnect is 649.35 Ω . To make the thermal resistance of a single RC thermal circuit unit at 20 Ω , the order of RC units is required to be about 30.

The interconnect structure shown in Fig. 2(a) is analyzed using the RC -TAN method. To balance accuracy and efficiency, a 30th-order RC thermal circuit is used to model the dielectric layer, and the remaining layers are modeled with a first-order RC thermal circuit.

Using the TAN method to analyze the equivalent thermal circuit, the impedance matrix Z_{mn} is obtained. The impedance matrix Z_{mn} can be transformed into a port impedance matrix which shown in (14) by adopting the method of (6)

$$Z_{mn} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}. \quad (14)$$

The ESD generated by the human body discharge mode (HBM) is used as the excitation of the interconnect signal layer. The current of HBM-ESD is denoted as follows:

$$I(t) = \frac{V_{ESD}}{R_{HBM}} \left[1 - \exp\left(-\frac{R_{HBM}}{L_p} t\right) \right] \cdot \exp\left(-\frac{t}{R_{HBM} C_{HBM}}\right) \quad (15)$$

where R_{HBM} , C_{HBM} , and L_p represent the discharge resistance (1500 Ω), the capacitance (100 pF) and the parasitic inductance (7.5 μ H) of human body.

The power generated on the interconnect line can be obtained as follows:

$$P(t) = I(t)^2 R \quad (16)$$

where R is impedance of interconnect.

The temperature response of the interconnect signal layer in the frequency domain can be obtained as follows:

$$T(s) = Z_{11}(s) \cdot P(s) \quad (17)$$

where $P(s)$ is the Laplace transformation of $P(t)$ and $Z_{11}(s)$ is the first element in matrix $Z(s)$ which is shown in (14).

An accurate solution of the temperature response in the time domain can be obtained by using the numerical Laplace inverse transform for $T(s)$.

Using our proposed RC -TAN method, the transient temperature response under the ESD pulse excitation is calculated, and the result is compared with that of FEM in the ANSYS platform as a comparison and verification, as illustrated in Fig. 7.

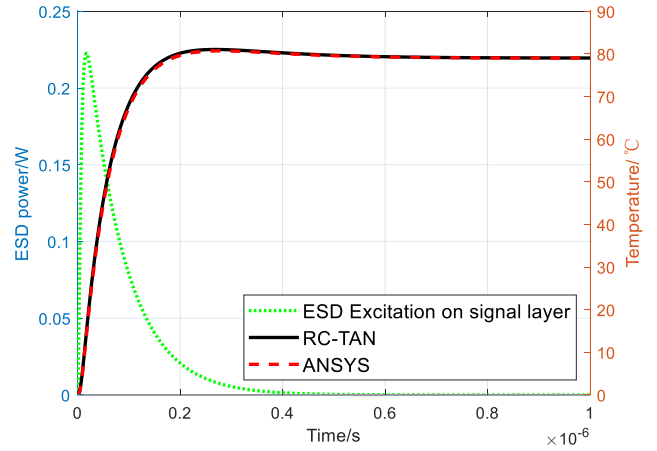


Fig. 7. Temperature response of signal interconnect to ESD pulses.

TABLE II
COMPARISON OF THERMAL RESPONSE AND RUNTIME

Method	ANSYS	RC-TAN	error
Maximum temperature/ °C	80.75	81.06	0.38%
Steady-state temperature/ °C	79.03	79.03	0%
Runtime/s	543	3	--

The simulation time of ANSYS is 543 s, while the simulation time of the proposed RC -TAN method is only 3 s. The maximum temperature error and the steady state error are only 0.38% and 0%, respectively, as shown in Table II.

The RC -TAN method is also applicable to the case of multiple signal layers and multiple excitations. In this article, the two-signal layer interconnect structure shown in Fig. 6 is used as an example, and the RC -TAN method is used to analyze its transient temperature response in the case of multiple excitations, where the ESD pulse is input on the signal layer 1. The temperature response on signal layer 2 is shown in Fig. 8(a). An ESD pulse is input on the signal layer 1, and after 400 ns, another ESD pulse is input on the signal layer 2. The temperature response on signal layer 1 is shown in Fig. 8(b). From the results, we can see that RC -TAN method has good performance in the calculation of multiexcitations and multisignal layer thermal coupling and the overall error does not exceed 2%.

In fact, multiple excitations can be input on any signal layer and the temperature response on any layer can be obtained by connecting the layer of interest to the port. The RC -TAN method can be extended for temperature response analysis in complex stacking structures and excitation environments, and it can save a great deal of the computation time compared to the FEM method.

V. ELECTROTHERMAL SENSITIVITY ANALYSIS OF PACKAGE INTERCONNECTS

In the design of package interconnects, there is an increasing interest in the factors that affect the temperature response. Batch simulations with simulation software to study and explore the laws of interconnect electrothermal response require not only considerable time, but also the support of high performance computing equipment. However, in the current

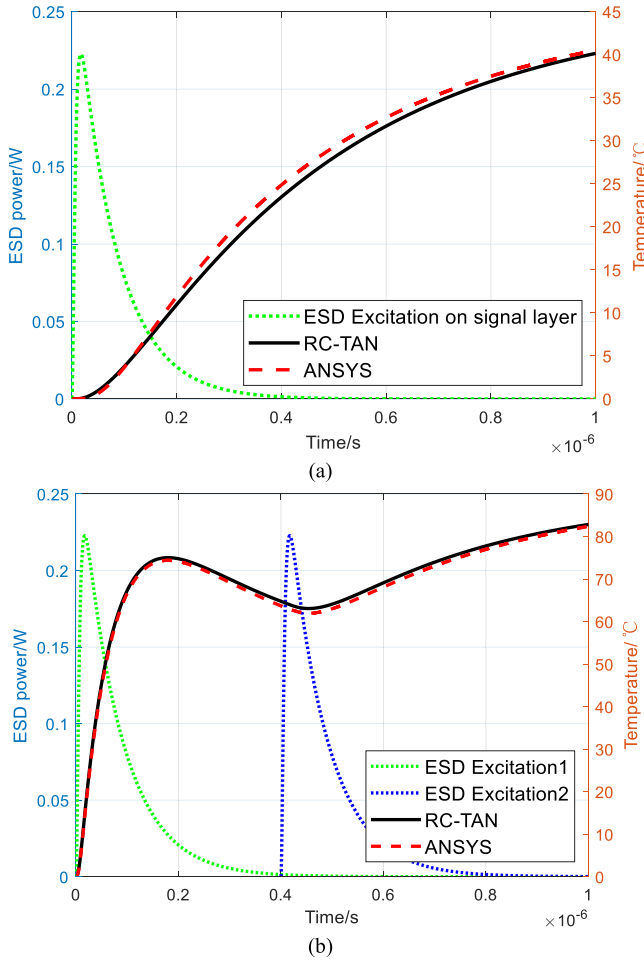


Fig. 8. Temperature response of dual interconnects under ESD pulses at different times. (a) Temperature response of signal layer 2 when ESD excitation is input on signal layer 1. (b) Temperature response of signal layer 1 when ESD excitation is input on signal layers 1 and 2 at different times respectively.

consumer electronics market, most engineers have to use their previous design experience for interconnect structures in order to reduce the time consumption of product design. Therefore, there is a lack of systematic simulation studies for the design of interconnect structures. The *RC-TAN* method solves this problem and enables efficient and fast electrothermal analysis of interconnect structures even with conventional computing equipment. It is also possible to conduct parameter sweep of the interconnect structure and perform sensitivity analysis of the parameters by using *RC-TAN* method.

A. Sensitivity Analysis of the Parameters of the Interconnect Structure

In order to investigate the law of the influence of the dimensional parameters of the package interconnect structure on the electrothermal response, the width and length of the interconnect structure as well as the thickness of the metal layer are scanned by using the *RC-TAN* method respectively. The simulation results are also compared with that of ANSYS to verify its accuracy.

First, the width and height of the interconnect structure are scanned from 1.0 to 2.0 μm and 100 to 200 μm , respectively,

TABLE III
SUMMARIZATION OF THE SENSITIVITY ANALYSIS

Parameter	Size	C_T	R_T	Temperature
W	$\uparrow$	$\uparrow$	$\downarrow$	$\downarrow$
L	$\uparrow$	$\uparrow$	$\downarrow$	$\downarrow$
h	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$

and the results are shown in Figs. 9 and 10. Second, the thickness of the metal layer is scanned from 0.5 to 1.0 μm , and the results are shown in Fig. 11, where L , W , and the metal layer h vary in the same proportion.

From the results, we can see that the peak and steady-state temperatures of the interconnects under ESD excitation decrease with the increase size of the interconnect structure. As shown in Figs. 9–11, the change of W and L make the temperature drop by 43° at most, while the change of h makes the temperature drop by 30° at most, so the temperature response of the interconnect structure is more sensitive to the length and width. In addition, the comparison with the results of ANSYS also shows the accuracy of the *RC-TAN* method. The summarization is listed in Table III.

By analyzing the *RC* thermal circuit model, we can know that the temperature is inversely proportional to the thermal capacity and is directly proportional to the thermal resistance. When h increases, C_T and R_T will increase, but the K of the metal layer is large enough so that the change of R_T is not sensitive to the temperature, which is the reason for the decrease of the temperature.

According to the summarization, we conclude as follows.

- 1) The larger the size the more it can reduce the response temperature and increase the reliability of the interconnect structure.
- 2) When the size change leads to an increase in thermal capacity or a decrease in thermal resistance, the thermal response temperature of the interconnect structure will decrease.
- 3) Due to the fact that the temperature response of the interconnect structure is more sensitive to the length and width, and wider and longer structure dimensions should be preferred when designing interconnect structures with high temperature reliability requirements.

B. Influence of Cascading Design on the Temperature Response

As mentioned above, highly integrated chips perform more functions in smaller areas, which requires a multilayer design in the package interconnect structure that allows multiple signals to be arranged in layers to transmit more signals.

In order to investigate the influence of the layer arrangement on the temperature response, the *RC-TAN* method was used to calculate the temperature response for the interconnect structure with three signal layers shown in Fig. 12(a), where the same excitation is input on the signal layers at different locations and their temperature responses are calculated. The ESD excitation is input on Metal Layer1 and Metal Layer2, and their temperature responses are calculated respectively,

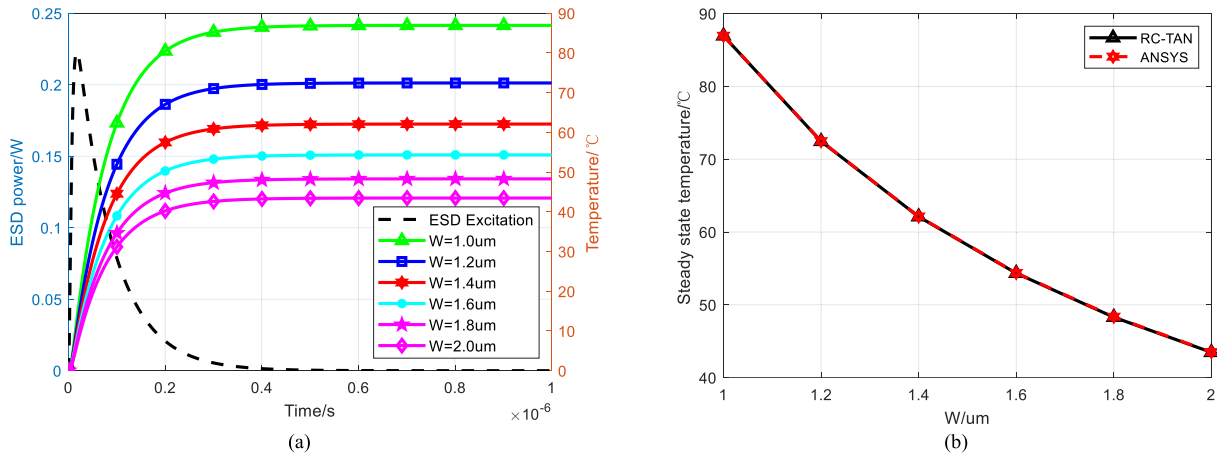


Fig. 9. Scanning results for the width of the interconnect structure. (a) Transient response. (b) Comparison of steady-state response with ANSYS results.

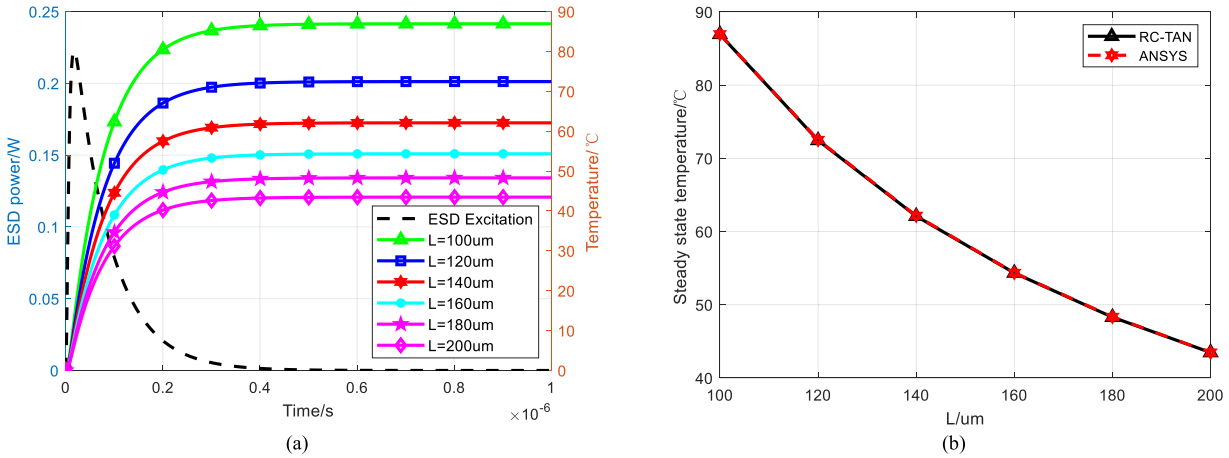


Fig. 10. Scanning results for the length of the interconnect structure. (a) Transient response. (b) Comparison of steady-state response with ANSYS results.

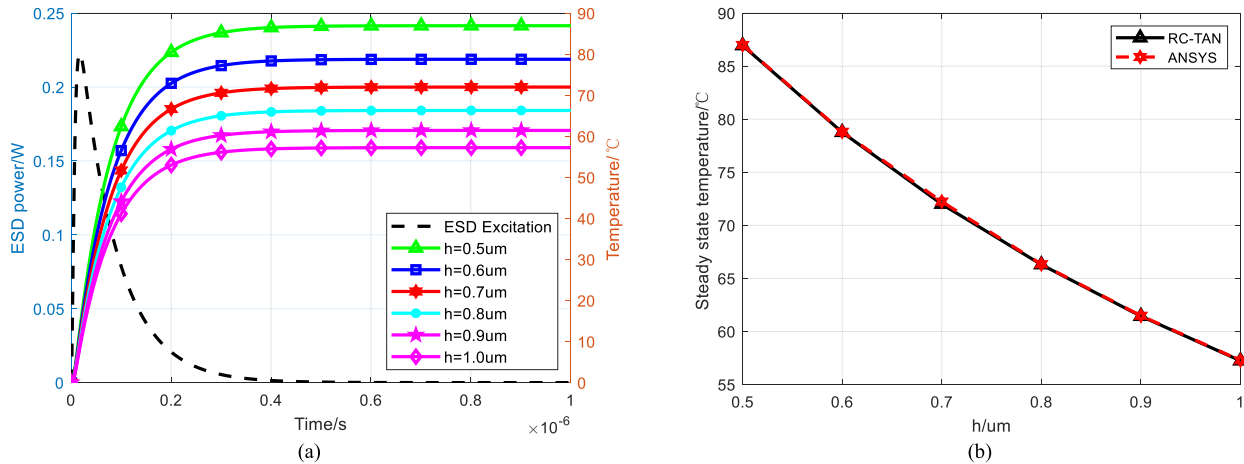


Fig. 11. Scanning results of metal layer thickness of interconnected structures. (a) Transient response. (b) Comparison of steady-state response with ANSYS results.

as shown in the Fig. 12(b). It can be seen that the peak temperature of the signal line in the middle layer is lower than that of the signal line in the outer layer. Based on the results, it can be concluded that when the interconnect structure is designed with multiple layers and there is no active heat dissipation, the temperature-sensitive signal lines should be placed in the middle layer because the signal lines in the middle layer have higher temperature reliability.

To sum up, this section demonstrates not only the fast modeling capability of the proposed RC-TAN method, but also its application in interconnect thermal design studies. By using the method to perform a series of effective and quantitative thermal calculations on interconnect structures, the thermal variation patterns with design parameters can be summarized. Based on these findings, effective interconnect thermal design guidelines can be developed to guide the design

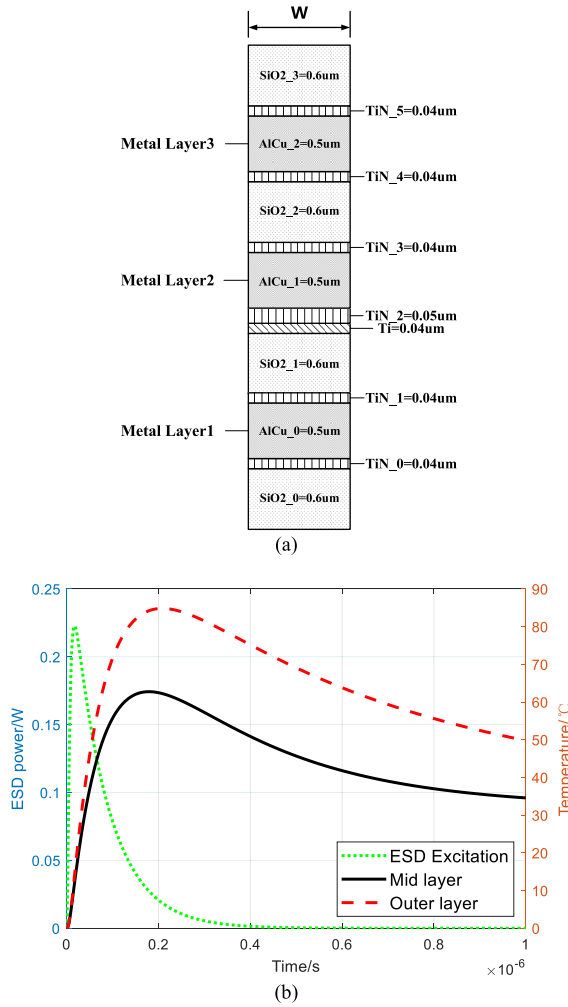


Fig. 12. Interconnect structure of the three signal layers. (a) Cascade of three signal layer interconnect structures. (b) Comparison of temperature response for each of signal layer 1 and signal layer 2 with ESD excitation.

of interconnect structures, thereby reducing the design time and realizing practical engineering implementation.

VI. CONCLUSION

In this article, the RC-TAN method is applied to the electrothermal analysis of package interconnects for the first time. This method can calculate the time domain temperature response of the interconnect quickly and accurately. Meanwhile, it is suitable for multilayer interconnect structures and can calculate the response results of multiple excitations simultaneously, which greatly saves time without sacrificing accuracy and improves the performance of thermal coupling for multilayer signal lines. The method is used to analyze the electrothermal sensitivity of the package interconnect, and the criteria which can guide the design of interconnect stacking structure are obtained: 1) interconnect structure with larger size has better reliability; 2) temperature response is more sensitive to the width and length of the line; and 3) interconnects placed in the middle layer have better reliability.

APPENDIX

Numerical inverse Laplace transform [35] enables a swift conversion from the frequency domain to the time domain,

which can be very advantageous in a wide range of academic and practical settings. The inverse Laplace transform is defined as

$$f(t) = L^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{\gamma-j\infty}^{\gamma+j\infty} F(s)e^{st} ds \quad (\text{A.1})$$

where the integration variable is $s = \gamma + j\omega$.

The transform $F(s)$ is supposed to fulfill the following assumptions:

- 1) $F(s)$ is regular for $\text{Re}\{s\} > 0$;
- 2) $\lim_{|s| \rightarrow \infty} F(s) = 0$;
- 3) $F^*(s) = F(s^*)$, where the asterisk means complex conjugate value.

To simplify the evaluation of the integral, we approximate the exponential function by the following equation:

$$\begin{aligned} e^{st} E_c(st, a) &= \frac{e^a}{2 \cosh(a - st)} \\ &= \frac{e^a}{e^a e^{-st} + e^{-a} e^{st}} = \frac{e^{st}}{1 + e^{-2a} e^{2st}}. \end{aligned} \quad (\text{A.2})$$

When $a > \gamma t$, then $|e^{-2a} e^{2st}| \ll 1$ and the fraction $1/(1 + e^{-2a} e^{2st})$ can be expanded as a convergent MacLaurin series, as shown in the following:

$$E_c(st, a) = e^{st} + \sum_{n=1}^{\infty} (-1)^n e^{-2na} e^{(2n+1)st}. \quad (\text{A.3})$$

The sum in (A.3) represents the error of approximation of the exponential function that can be arbitrarily suppressed by a choice of the parameter a . Then

$$\begin{aligned} f(t) f_c(t, a) &= \frac{1}{2\pi j} \int_{\gamma-j\infty}^{\gamma+j\infty} F(s) E_c(st, a) ds \\ &= \frac{1}{2\pi j} \int_{\gamma-j\infty}^{\gamma+j\infty} F(s) e^{st} ds + \frac{1}{2\pi j} \\ &\quad \times \int_{\gamma-j\infty}^{\gamma+j\infty} F(s) \sum_{n=1}^{\infty} (-1)^n e^{-2na} e^{(2n+1)st} ds \\ &= f(t) + \sum_{n=1}^{\infty} (-1)^n e^{-2na} f[(2n+1)t] \\ &= f(t) - e^{-2a} f(3t) + e^{-4a} f(5t) - \dots \\ &= f(t) + e_c(t, a). \end{aligned} \quad (\text{A.4})$$

Also, the error $e_c(t, a)$ of the inverse transform can be kept at low values by appropriate choice of the parameter a . The function $E_c(st, a)$ can be expressed as an infinite sum of rational functions. The hyperbolic cosine function can be expressed as the following equation [33]:

$$\frac{1}{\cosh z} = 2\pi \sum_{n=0}^{\infty} \frac{(-1)^n (n + 1/2)}{(n + 1/2)^2 \pi^2 + z^2}. \quad (\text{A.5})$$

Substituting equating (A.5) into (A.2), $E_c(st, a)$ can be simplified as

$$E_c(st, a) = \pi e^a \sum_{n=0}^{\infty} \frac{(-1)^n (n + 1/2)}{(n + 1/2)^2 \pi^2 + (a - st)^2}. \quad (\text{A.6})$$

The function $f_c(t, a)$ can be rewritten as

$$f_c(t, a) = \frac{e^a}{2j} \int_{\gamma-j\infty}^{\gamma+j\infty} F(s) \sum_{n=0}^{\infty} \frac{(-1)^n (n+1/2)}{(n+1/2)^2 \pi^2 + (a-st)^2} ds. \quad (\text{A.7})$$

By interchanging the sequence of summation and integration, we get the following equation:

$$f_c(t, a) = \frac{e^a}{2j} \sum_{n=0}^{\infty} (-1)^n (n+1/2) I_n \quad (\text{A.8})$$

where I_n can be expressed as

$$\begin{aligned} I_n &= \int_{\gamma-j\infty}^{\gamma+j\infty} \frac{F(s)}{(n+1/2)^2 \pi^2 + (a-st)^2} ds \\ &= \int_{\gamma-j\infty}^{\gamma+j\infty} \frac{F(s)}{G_n(s)} ds = \int_{\gamma-j\infty}^{\gamma+j\infty} H_n(s) ds. \end{aligned} \quad (\text{A.9})$$

To evaluate the integral, we complete the integration path by a semicircle with the infinite radius. Due to assumption 2, the function $F(s)$ equals zero along this semicircle. The integral I_n is then equal to the sum of the residua of $H_n(s)$ in the poles lying inside the integration contour multiplied by $2\pi j$. The poles of $H_n(s)$ in turn are the poles of $F(s)$ (these are concentrated in $\text{Re}\{s\} \leq 0$) plus the roots of the equation $G_n(s) = 0$, they can be expressed as follows:

$$s_{1,2} = \frac{a \pm j(n+1/2)\pi}{t} \quad (\text{A.10})$$

which lie in the right half-plane, to the right from the original path of integration $\text{Re}\{s\} = \gamma$. It is therefore advantageous to close the path of integration here and to calculate the I_n , as shown in the following:

$$\begin{aligned} I_n &= -2\pi j \left[\frac{F(s_1)}{2\pi j t (n+1/2)} - \frac{F(s_2)}{2\pi j t (n+1/2)} \right] \\ &= \frac{F(s_1^*) - F(s_1)}{t(n+1/2)} = \frac{F^*(s_1) - F(s_1)}{t(n+1/2)} = \frac{-2j\text{Im}\{F(s_1)\}}{t(n+1/2)} \\ &= \frac{-2j\text{Im}\{F[a + j(n+1/2)\pi]/t\}}{t(n+1/2)}. \end{aligned} \quad (\text{A.11})$$

Finally, the $f_c(t, a)$ can be calculated as follows:

$$f_c(t, a) = -\frac{e^a}{t} \sum_{n=0}^{\infty} (-1)^n \text{Im} \left\{ F \left[\frac{a}{t} + j(n+1/2) \frac{\pi}{t} \right] \right\}. \quad (\text{A.12})$$

REFERENCES

- [1] R. K. Ulrich and W. D. Brown, "Introduction and overview of microelectronic packaging," *IEEE Trans. Adv. Electron. Packag.*, vol. 29, no. 1, pp. 1–27, Feb. 2006.
- [2] N. Khan et al., "Development of 3-D silicon module with TSV for system in packaging," *IEEE Trans. Compon. Packag., Technol.*, vol. 33, no. 1, pp. 3–9, Mar. 2010.
- [3] S. F. Al-Sarawi, D. Abbott, and P. D. Franzone, "A review of 3-D packaging technology," *IEEE Trans. Compon., Packag., Manuf. Technol. B*, vol. 21, no. 1, pp. 2–14, Feb. 1998.
- [4] X. Fan, "Wafer level packaging (WLP): Fan-in, fan-out and three-dimensional integration," in *Proc. 11th Int. Thermal, Mech. Multi-Phys. Simul., Exp. Microelectron. Microsystems*, 2010, pp. 1–7.
- [5] R. R. Tummala, "Packaging: Past, present and future," in *Proc. 6th Int. Conf. Electron. Packag. Technol.*, 2005, pp. 3–7.
- [6] C. Wyland and W. Nunn, "High density package interconnect wire bond strip line and method therefor," U.S. Patent WO04/05 398 7P1, Dec. 4, 2003.
- [7] R. R. Khan and S. Z. Zhao, "Interconnect structure and formation for package stacking of molded plastic area array package," U.S. Patent US771 445 3B2, Feb. 11, 2007.
- [8] J. A. Davis et al., "Interconnect limits on gigascale integration (GSI) in the 21st century," *Proc. IEEE*, vol. 89, no. 3, pp. 305–324, Mar. 2001.
- [9] K. Banerjee and A. Mehrotra, "Global (interconnect) warming," *IEEE Circuits Devices Mag.*, vol. 17, no. 5, pp. 16–32, Sep. 2001.
- [10] V. Szekely, M. Rencz, and B. Courtois, "A step forward in the transient thermal characterization of chips and packages," *Microelectron. Rel.*, vol. 39, no. 1, pp. 89–96, Jan. 1999.
- [11] X. Gui, P. W. Webb, and G. B. Gao, "Use of the three-dimensional TLM method in the thermal simulation and design of semiconductor devices," *IEEE Trans. Electron Devices*, vol. 39, no. 6, pp. 1295–1302, Jun. 1992.
- [12] S. Rzepka, K. Banerjee, E. Meusel, and C. Hu, "Characterization of self-heating in advanced VLSI interconnect lines based on thermal finite element simulation," *IEEE Trans. Compon., Packag., Manuf. Technol.*, A, vol. 21, no. 3, pp. 406–411, Sep. 1998.
- [13] J. Zhang, M. O. Bloomfield, J.-Q. Lu, R. J. Gutmann, and T. S. Cale, "Modeling thermal stresses in 3-D IC interwafer interconnects," *IEEE Trans. Semiconductor Manuf.*, vol. 19, no. 4, pp. 437–448, Nov. 2006.
- [14] M. Pedram and S. Nazarian, "Thermal modeling, analysis, and management in VLSI circuits: Principles and methods," *Proc. IEEE*, vol. 94, no. 8, pp. 1487–1501, Aug. 2006.
- [15] Y.-B. Shi, W.-Y. Yin, J.-F. Mao, P. Liu, and Q. H. Liu, "Transient electrothermal analysis of multilevel interconnects in the presence of ESD pulses using the nonlinear time-domain finite-element method," *IEEE Trans. Electromagn. Compat.*, vol. 51, no. 3, pp. 774–783, Aug. 2009.
- [16] L. J. Ladani, "Numerical analysis of thermo-mechanical reliability of through silicon vias (TSVs) and solder interconnects in 3-dimensional integrated circuits," *Microelectron. Eng.*, vol. 87, no. 2, pp. 208–215, Feb. 2010.
- [17] Y. Dong, M. Tang, and J. Mao, "Transient electrothermal analysis of on-chip interconnects in the presence of ESD pulses using DGTDM method," in *Proc. Int. Appl. Comput. Electromagn. Soc. Symp. China (ACES)*, 2018, pp. 1–2.
- [18] B. Li, M. Tang, H. Yue, Y. Tang, and J. Mao, "Efficient transient thermal simulation of ICs and packages with Laguerre-based finite-element method," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 10, no. 2, pp. 203–211, Feb. 2020.
- [19] R. Ranade, H. He, J. Pathak, N. Chang, A. Kumar, and J. Wen, "A thermal machine learning solver for chip simulation," in *Proc. ACM/IEEE 4th Workshop Mach. Learn. CAD (MLCAD)*, Salt Lake City, UT, USA, Sep. 2022, pp. 111–117.
- [20] K. Sikka, R. Lall, and T. Sinha, "Artificial neural networks for package thermal analysis," in *Proc. 20th IEEE Intersociety Conf. Thermal Thermomechanical Phenomena Electron. Syst. (iTherm)*, San Jose, CA, USA, Jun. 2021, pp. 103–113.
- [21] J. Wen et al., "DNN-based fast static on-chip thermal solver," in *Proc. 36th Semiconductor Thermal Meas., Model. Manage. Symp. (SEMI-THERM)*, San Jose, CA, USA, Mar. 2020, pp. 65–75.
- [22] L. Chen, W. Jin, and S. X.-D. Tan, "Fast thermal analysis for chiplet design based on graph convolution networks," in *Proc. 27th Asia South Pacific Design Autom. Conf. (ASP-DAC)*, Salt Lake City, UT, USA, Jan. 2022, pp. 485–492.
- [23] A. Sridhar, A. Vincenzi, M. Ruggiero, and D. Atienza, "Neural network-based thermal simulation of integrated circuits on GPUs," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 31, no. 1, pp. 23–36, Jan. 2012.
- [24] P. Salome, C. Leroux, P. Crevel, and J. P. Chante, "Investigations on the thermal behavior of interconnects under ESD transients using a simplified thermal RC network," *Microelectron. Rel.*, vol. 39, no. 11, pp. 1579–1591, Nov. 1999.
- [25] K. M. Amin, W. Fikry, M. N. Sabry, and H. F. Ragai, "Modelling of thermal failure of metallic interconnects under electrostatic discharge transients," *Electron. Lett.*, vol. 41, no. 19, pp. 1056–1057, Sep. 2005.
- [26] A. Jain, R. E. Jones, R. Chatterjee, and S. Pozder, "Analytical and numerical modeling of the thermal performance of three-dimensional integrated circuits," *IEEE Trans. Compon. Packag., Technol.*, vol. 33, no. 1, pp. 56–63, Mar. 2010.

- [27] Q. Shang, X. Li, and J. Mao, "Transient thermal analysis of global interconnects based on transmission lines," in *Proc. Asia-Pacific Symp. Electromagn. Compat.*, 2012, pp. 245–248.
- [28] G. Kron, *Tensor Analysis of Networks*. New York, NY, USA: Wiley, 1939.
- [29] G. Kron, "General asymmetrical networks," in *Tensors for Circuits*. New York, NY, USA: Dover, 1959, pp. 89–104.
- [30] O. Maurice, "Theoretical application of the tensorial analysis of network for EMC at the system level," in *Proc. Int. Workshop Electromagn. Compat. Integr. Circuits (EMC Compo)*, 2007, pp. 1–10.
- [31] S. Leman, B. Demoulin, and O. Maurice, "Simple example for testing the tensorial analysis of electrical networks applied to EMC," in *Proc. Symp. Embedded (EMC)*, Saint Etienne du Rouvray, France, 2007, p. 4C.
- [32] O. Maurice, *Elements of Theory for Electromagnetic Compatibility and Systems*. Aix-en-Provence, France: Bookelis, 2017.
- [33] S. H. Voldman, "A review of electrostatic discharge (ESD) in advanced semiconductor technology," *Microelectron. Rel.*, vol. 44, no. 1, pp. 33–46, Jan. 2004.
- [34] Z.-F. Xu and B. Ravelo, *Tensorial Analysis of Networks (TAN) Modelling for PCB Signal Integrity and EMC Analysis*. London, U.K.: The Institution of Engineering and Technology, 2020.
- [35] J. Valsa and L. Brančik, "Approximate formulae for numerical inversion of Laplace transforms," *Int. J. Numer. Model., Electron. Netw., Devices Fields*, vol. 11, no. 3, pp. 153–166, May 1998.



Yang Liu received the B.S. degree in signal processing and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2001 and 2007, respectively.

He was a Visiting Scholar with the Department of Computer Science and Engineering, University of California at San Diego, San Diego, CA, USA, from 2012 to 2013. He joined Xidian University in 2006, where he is currently an Associate Professor with the Department of Electronic Engineering. His current research interests include power

integrity modeling and design, and modeling and simulation of high-speed interconnects.

Dr. Liu was a recipient of the Science and Technology Development Award of Shannxi Province in 2012.



Dehang Wu (Student Member, IEEE) received the B.S. degree in electronic information engineering from Xidian University, Xi'an, China, in 2020, where he is currently pursuing the master's degree with the Key Laboratory of High Speed Circuit Design and EMC of Ministry of Education.

His current research interests include power integrity modeling and design, and modeling and simulation of high-speed interconnects.



Zhifei Xu (Member, IEEE), received the master's degree in microelectronics from the École Supérieure d'Ingénieurs en Électrotechnique et Électronique, Paris, France, in 2016, and the Ph.D. degree in electrical engineering from the Université de Rouen Normandie, Rouen, France, in 2019.

He was a Post-Doctoral Researcher at the Electromagnetic Compatibility (EMC) Laboratory, Missouri University of Science and Technology, Rolla, MO, USA, from 2019 to the end of 2020. He was also a Senior Signal/Power Integrity Engineer in Kandou

Bus, Switzerland. He is currently the Director of high-speed platform at Ningbo Detool Technology Company Ltd., Ningbo, China. He is also one of the contributors to the applications of Kron's model. His research area covers signal/power integrity and electromagnetic compatibility (EMC) analysis in high-speed interconnect systems, hardware security, RFI.



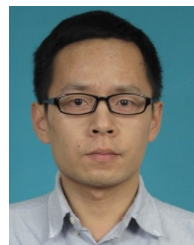
Xiuqin Chu (Senior Member, IEEE) received the B.S. degree in electronic engineering from Xi'an Shiyou University, Xi'an, China, in 1994, and the M.S. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, in 1997 and 2003, respectively.

In April 1997, she joined Xidian University and is currently a Professor and the Ph.D. Candidate Advisor with the School of Electronic Engineering. From 2016 to 2017, she was a Visiting Scholar with the EMC Laboratory, Missouri University of Science and Technology, Rolla, MO, USA. Her current research interests include signal/power integrity, modeling for high-speed links, and jitter analysis in high-speed digital systems.



Jun Wang (Member, IEEE) received the B.S. degree in electronic information science and technology from the Langfang Teachers College, Langfang, China, in 2008, and the Ph.D. degree from Xidian University, Xi'an, China, in 2017.

From 2019 to 2020, she was a Visiting Scholar with EMC Laboratory, Missouri University of Science and Technology, Rolla, MO, USA. She is currently a Lecturer with the School of Electronic Engineering, Xidian University. Her current research interests focus on signal integrity, jitter, and power integrity.



Kai-Da Xu (Senior Member, IEEE) received the B.E. and Ph.D. degrees in electromagnetic field and microwave technology from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2009 and 2015, respectively.

From 2012 to 2014, he was a Visiting Researcher with the Department of Electrical and Computer Engineering, Duke University, Durham, NC, USA, under the financial support from the China Scholarship Council. In 2015, he joined the Department of Electronic Science, Xiamen University, Xiamen,

China, as an Assistant Professor. From 2016 to 2017, he was a Postdoctoral Fellow with the State Key Laboratory of Millimeter Waves, City University of Hong Kong, Hong Kong. From 2018 to 2019, he was an Honorary Fellow with the Department of Electrical and Computer Engineering, University of Wisconsin–Madison, Madison, WI, USA. He was successfully selected into the "Youth Talent Support Program" of Xi'an Jiaotong University (XJTU) in 2019, where he joined the School of Information and Communications Engineering in 2020. He has authored and coauthored over 170 papers in peer-reviewed journals and over 50 papers in conference proceedings. His current research interests include RF/microwave/millimeter-wave circuits and antenna arrays.

Dr. Xu was the Japan Society for the Promotion of Science (JSPS) Fellow with the Department of Communications Engineering, Graduate School of Engineering, Tohoku University, from 2019 to 2021. He received a fellowship from the JSPS. He received the UESTC Outstanding Graduate Awards in 2009 and 2015, respectively. He was a recipient of the National Graduate Student Scholarship in 2012, 2013, and 2014, from the Ministry of Education, China. He has been serving as an Editorial Board Member for both the *AEÜ-International Journal of Electronics and Communications* and *MDPI Electronics*. Also, he has been serving as an Associate Editor for the *IEEE ACCESS*, *IET Microwaves, Antennas and Propagation*, and *Electronics Letters*.